Inflation Buoyancy Sector Lagrangian with Stationarity Constraint

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PAPER_1089: Inflation Buoyancy Sector Lagrangian with Stationarity Constraint

Abstract

We construct the Lagrangian for the inflation buoyancy sector of the UQFF framework:

$$\mathcal{L}_{\text{infl}} = -\beta_i \cdot U_g \cdot \Omega \cdot \frac{M}{d} \cdot [\text{UA}] + F_n \cdot \Phi$$

The action $S = \int \mathcal{L}_{\text{infl}}\, dt$ is varied with respect to the inflaton field ϕ_{infl} to obtain the stationarity condition $\delta S/\delta\phi_{\text{infl}} = 0$. The Lagrangian decomposes into a gravitational potential term and a neutron-phonon driving term, whose balance determines the inflationary slow-roll dynamics.

§1 Lagrangian Construction

§1.1 Gravity Term

$$\mathcal{L}_{\text{grav}} = -\beta_i \cdot U_g \cdot \Omega \cdot \frac{M}{d} \cdot [\text{UA}]$$

with:

- $\beta_i = 0.603$ (buoyancy coupling)
- $U_g = \mu_s \cdot \nabla(M_s/r)$ (gravitational potential energy)
- Ω = angular rotation frequency
- M/d = mass-to-distance linear density
- $[\text{UA}] = 10^{-4}$ (universal abundance parameter)

§1.2 Neutron-Phonon Term

$$\mathcal{L}_{\text{neutron}} = F_n \cdot \Phi$$

with $F_n = 10^{-10}$ N (neutron exchange force) and $\Phi = \Phi_0 \cdot S_{26}$ (phonon amplitude).

§1.3 Total Lagrangian

$$\mathcal{L}_{\text{infl}} = \mathcal{L}_{\text{grav}} + \mathcal{L}_{\text{neutron}}$$

§2 Stationarity and Euler-Lagrange Equation

The action principle requires:

$$\frac{\delta S}{\delta \phi_{\text{infl}}} = \frac{\partial \mathcal{L}}{\partial \phi} - \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{\phi}} = 0$$

Since $\mathcal{L}_{\text{infl}}$ depends on ϕ through $U_g(\phi)$ and $\Phi(\phi)$:

$$-\beta_i \Omega \frac{M}{d}[\text{UA}]\frac{\partial U_g}{\partial \phi} + F_n \frac{\partial \Phi}{\partial \phi} = 0$$

§3 Gravity vs Neutron Balance

At stationarity, the ratio:

$$\left|\frac{\mathcal{L}_{\text{grav}}}{\mathcal{L}_{\text{neutron}}}\right| = \frac{\beta_i \cdot U_g \cdot \Omega \cdot M \cdot [\text{UA}]}{d \cdot F_n \cdot \Phi}$$

determines whether inflation is gravity-dominated (> 1) or phonon-driven (< 1).

§4 Benchmark: Solar System

Parameter	Value	Source
M	1.989×10^{30} kg	Solar mass
$r = d$	1.496×10^{11} m	1 AU
Ω	2.87×10^{-6} rad/s	Solar rotation
U_g	8.87×10^{29} J	$GM_\odot/r$
$\mathcal{L}_{\text{grav}}$	-2.06×10^{8} J	Gravity sector
$\mathcal{L}_{\text{neutron}}$	1.86×10^{11} J	Neutron-phonon
Ratio	1.11×10^{-3}	Phonon-dominated

§5 Physical Interpretation

The inflation buoyancy sector is phonon-dominated under solar conditions, meaning the SCm vacuum lattice oscillation provides the dominant energy injection for inflationary expansion. The gravitational restoring term acts as a regulator preventing runaway inflation.

References

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